1. Lab 12: Wireless LANs (WLAN) - WiFi and 802.11 Analysis

Lab Name: Lab 12: Wireless LANs (WLAN) - WiFi and 802.11 Analysis **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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AI Policy & Acceptable Use Guidelines

 AI Policy & Acceptable Use Guidelines - Yellow Icon

Artificial Intelligence tools can be powerful learning aids, but they must be used ethically and responsibly to ensure you develop core networking skills.

- **Lab Assignments:** Do not use AI to perform lab steps or write your lab submissions. However, if you get stuck on a concept or technical obstacle, you may use AI as a virtual tutor to ask clarifying questions and help you troubleshoot.

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2. Background

Wireless Local Area Networks (WLANs) operate under the **IEEE 802.11 (WiFi)** standard. Key concepts include:

1. **IEEE 802.11 Frame Format**: Unlike Ethernet frames which use a 2-address format (Source and Destination), 802.11 frames use up to a **4-address format**. This is because frames are transmitted over a shared wireless medium to an intermediate **Access Point (AP)**, which acts as a bridge to the wired distribution system.
2. **Addressing Mapping**: The roles of the address fields (Address 1, Address 2, Address 3, Address 4) are determined by the `To DS` (To Distribution System) and `From DS` flags in the Frame Control field:
   - `To DS = 0, From DS = 1` (Downlink from AP to Client): Address 1 is Destination (Client), Address 2 is BSSID (AP), Address 3 is Source (Sender).
   - `To DS = 1, From DS = 0` (Uplink from Client to AP): Address 1 is BSSID (AP), Address 2 is Source (Client), Address 3 is Destination (Receiver).
3. **802.11 Frame Types**: Frames are classified into:
   - **Management Frames**: Beacons, probe requests/responses, association requests/responses. (Beacons are broadcast periodically by the AP to advertise the network SSID, rates, and capabilities).
   - **Control Frames**: RTS/CTS (Request/Clear to Send) to avoid collisions and solve the hidden terminal problem, and ACKs to confirm frame delivery.
   - **Data Frames**: Encapsulate the actual network layer payloads.
4. **WPA2 4-Way Handshake**: Wireless links are encrypted to prevent eavesdropping. When a client associates, it performs a **4-way cryptographic handshake** using EAPOL (EAP over LAN) key exchanges to verify the Pre-Shared Key (PSK) and generate temporal encryption keys (PTK and GTK).

In this lab, you will deploy a wireless network in CML using a WiFi Access Point and two WiFi Client nodes, perform packet captures on the wireless interface, and analyze management, control, data, and security handshake sequences.

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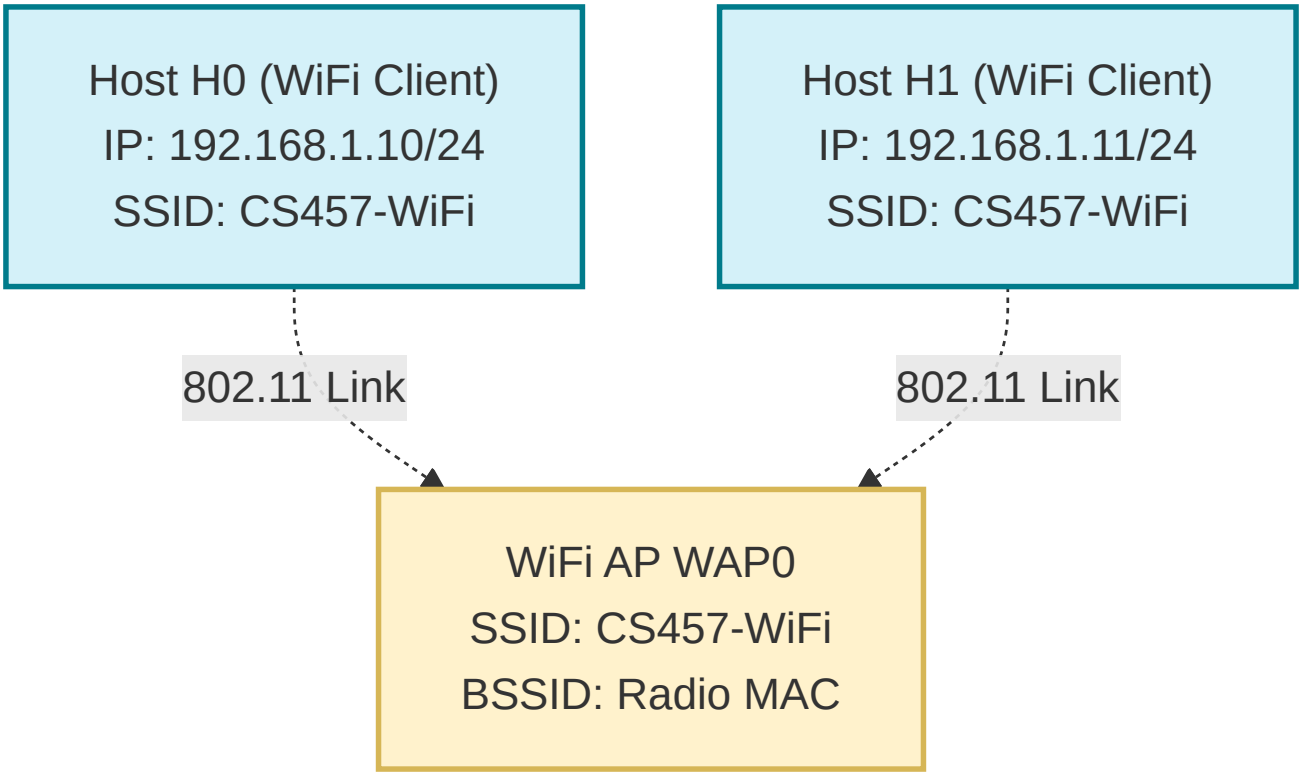
3. Objective / Purpose

To configure a basic WLAN infrastructure in CML, capture and analyze 802.11 management frames (beacons), inspect 802.11 frame address fields and control flag maps, and analyze the WPA2 4-way cryptographic key exchange in

Wireshark.

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4. Topology Diagram



Details:

- 1x WiFi Access Point (`<initials>-WAP0`) broadcasting SSID `CS457-WiFi` .
- 2x WiFi Clients (`<initials>-H0` and `<initials>-H1`) associated with the AP.

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 2x WiFi Client Nodes, 1x WiFi Access Point Node

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6. Configuration / Methodology

WLAN Configuration Table

Device	CML Node Name	Interface	IP Address	SSID	Security Mode	Pre-Shared Key
Access Point	`<initials>-WAP0`	Radio0	192.168.1.1/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`

Client 0	`<initials>-H0`	wlan0	192.168.1.10/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`
Client 1	`<initials>-H1`	wlan0	192.168.1.11/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`

Step-by-Step Configuration Guide

Foreground vs. Background Daemon Execution

When configuring wireless nodes in Linux, wireless daemons (`hostapd` on APs and `wpa_supplicant` on Clients) can be executed in two modes:

- **Background Mode (`-B` flag)**: Runs the daemon silently in the background, freeing up your terminal prompt.
  - Start AP in background: `hostapd -B /home/cs457/hostapd.conf`
  - Start Client in background: `wpa_supplicant -B -i wlan0 -c /tmp/wpa_supplicant.conf`
- **Foreground Debug Mode (Omit `-B` , add `-d` or `-dd`)**: Prints real-time association events, probe requests, and EAPOL key handshakes directly to your terminal. Useful for troubleshooting.
  - Run AP in foreground debug mode: `pkill hostapd && hostapd -dd /home/cs457/hostapd.conf`
  - Run Client in foreground debug mode: `pkill wpa_supplicant && wpa_supplicant -dd -i wlan0 -c /tmp/wpa_supplicant.conf`
  - *(Press `Ctrl+C` to terminate a foreground daemon).*

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1. Access Point Setup (`<initials>-WAP0`)

In CML, open the **Edit Config** tab of the WiFi AP node (`KH-WAP0`) and paste the following `#cloud-config` :

```
#cloud-config
hostname: KH-WAP0
manage_etc_hosts: True
system_info:
  default_user:
    name: cs457
password: cs457
chpasswd:
  expire: False
ssh_pwauth: True
ssh_authorized_keys:
  - your-ssh-pubkey-line-goes-here
write_files:
  - path: /home/cs457/hostapd.conf
    content: |
      interface=wlan0
      driver=nl80211
      ssid=CS457-WiFi
      channel=6
      auth_algs=1
      wpa=2
      wpa_key_mgmt=WPA-PSK
      wpa_passphrase=wirelesspass123
```

```
      rsn_pairwise=CCMP
      beacon_int=1000
    owner: cs457:cs457
    permissions: '0644'
runcmd:
  - ip addr add 192.168.1.1/24 dev wlan0
  - ip link set wlan0 up
  - hostapd -B /home/cs457/hostapd.conf
```

2. Client Host Setup (`<initials>-H0` and `<initials>-H1`)

Standard Linux nodes do not support WPA2-PSK via legacy `iwconfig` (which only supports WEP). To connect to a WPA2-PSK network, use `wpa_supplicant` with `p2p_disabled=1` and `psk="wirelesspass123"` :

Option A: Interactive CLI on Client Nodes

1. **Configure Host `<initials>-H0`** :

```
# Bring up interface and assign IP
ip addr add 192.168.1.10/24 dev wlan0
ip link set wlan0 up

# Create WPA2 PSK configuration file directly with ctrl_interface enabled for wpa_cli
cat << 'EOF' > /tmp/wpa_supplicant.conf
ctrl_interface=/var/run/wpa_supplicant
p2p_disabled=1
network={
    ssid="CS457-WiFi"
    psk="wirelesspass123"
    key_mgmt=WPA-PSK
}
EOF

# Kill any previous instance and start wpa_supplicant daemon
pkill wpa_supplicant
wpa_supplicant -B -i wlan0 -c /tmp/wpa_supplicant.conf
```

2. **Configure Host `<initials>-H1`** :

```
# Bring up interface and assign IP
ip addr add 192.168.1.11/24 dev wlan0
ip link set wlan0 up

# Create WPA2 PSK configuration file directly
cat << 'EOF' > /tmp/wpa_supplicant.conf
ctrl_interface=/var/run/wpa_supplicant
p2p_disabled=1
network={
    ssid="CS457-WiFi"
    psk="wirelesspass123"
    key_mgmt=WPA-PSK
}
```

```
EOF

# Kill any previous instance and start wpa_supplicant daemon
pkill wpa_supplicant
wpa_supplicant -B -i wlan0 -c /tmp/wpa_supplicant.conf
```

Option B: Client `#cloud-config` (in CML Edit Config tab for Clients)

```
#cloud-config
hostname: KH-H0
manage_etc_hosts: True
system_info:
  default_user:
    name: cs457
password: cs457
chpasswd:
  expire: False
ssh_pwauth: True
write_files:
  - path: /etc/wpa_supplicant/wpa_supplicant.conf
    content: |
      ctrl_interface=/var/run/wpa_supplicant
      p2p_disabled=1
      network={
          ssid="CS457-WiFi"
          psk="wirelesspass123"
          key_mgmt=WPA-PSK
      }
    owner: root:root
    permissions: '0644'
runcmd:
  - ip addr add 192.168.1.10/24 dev wlan0
  - ip link set wlan0 up
  - wpa_supplicant -B -i wlan0 -c /etc/wpa_supplicant/wpa_supplicant.conf
```

(For Host H1, adjust hostname to `KH-H1` and IP to `192.168.1.11/24`).

Step-by-Step Traffic Generation & Execution:

> *[!IMPORTANT]* ***REMINDER: All screenshots, console outputs, and packet captures (.pcap files) must be taken from the SAME active lab session. Mixing screenshots or trace files across different runs or sessions is not permitted.***

1. **Start Packet Capture in CML GUI**:
   - Right-click the Access Point (`<initials>-WAP0`) or Client (`<initials>-H0`) node.
   - Select **Packet Capture**, choose **`Wireless`** (which captures on the `wlan0` interface), and click **Start**. *(Note: Selecting **Wireless** ensures raw 802.11 frames—beacons, 4-address headers, and WPA2 EAPOL handshakes—are recorded).*
2. **Verify Client Association**: Open Host H0's console and verify connection status using native built-in tools (`iw` or `wpa_cli`):

```
# Method 1: Using iw (standard modern netlink tool):
iw dev wlan0 link

# Method 2: Using wpa_cli status (from wpa_supplicant):
wpa_cli status
```

(Confirm that `SSID: CS457-WiFi` and the Access Point MAC address/BSSID are displayed, indicating successful association).

3. **Generate Traffic**: Ping Client H1 from Client H0:

```
ping -c 4 192.168.1.11
```

4. **Trigger & Log 4-Way Security Handshake**: To capture the complete WPA2 4-Way Handshake log output directly on the Access Point console: a. On AP node (`<initials>-WAP0`), stop background daemon and start `hostapd` in foreground debug mode:

```
pkill hostapd
hostapd -dd /home/cs457/hostapd.conf
```

b. On Client H1 console, trigger re-association:

```
wpa_cli -i wlan0 reassociate
```

c. Observe the live log stream on `<initials>-WAP0` displaying:

- `sending 1/4 msg of 4-Way Handshake`
- `received EAPOL-Key frame (2/4 Pairwise)`
- `sending 3/4 msg of 4-Way Handshake`
- `received EAPOL-Key frame (4/4 Pairwise)`
- `EAPOL-4WAY-HS-COMPLETED` d. Take a screenshot of this terminal window for your lab submission.

5. **Download Capture File from CML**:

- In the CML interface **Packet Capture** tab, click **Stop**.
- Click **Download** to save the `.pcap` file directly to your local computer with one click, then open it in Wireshark.

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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark and answer the following questions:

Part A: 802.11 Management Frames (Beacons)

1. Locate a **Beacon frame** sent by the AP (`<initials>-WAP0`).
   - What is the Source MAC address (the transmitter)?
   - What is the BSSID of the wireless network? Does this match the AP's MAC?
2. Inspect the **Beacon parameters** in the packet details pane:
   - What SSID is advertised?

- What WPA/WPA2 security requirements (e.g. Robust Secure Network - RSN group/pairwise cipher suites) are listed in the parameters?

Part B: 802.11 Frame Addressing & Control Flags

3. Locate an **802.11 Data frame** containing your ping traffic (ICMP request) sent from Client H0 to Client H1.
   - Expand the **Frame Control** field in the IEEE 802.11 header. What are the binary/hex values of the `To DS` and `From DS` flags?
   - Based on these flag states, why are these values set (what is the directional path of the packet)?
4. Inspect the address fields in the header of this data frame:
   - What is the MAC address value for:
     1. Address 1 (Receiver Address / RA)
     2. Address 2 (Transmitter Address / TA)
     3. Address 3 (Source/Destination Address depending on DS flags)
   - Map these addresses to the actual CML nodes in your topology (H0, H1, or WAP0).
5. Explain why an IEEE 802.11 wireless frame header requires up to **four** MAC address fields, whereas a standard Ethernet II wired frame header only requires **two**. (Discuss the bridging role of the AP).

Part C: WPA2 Security and 4-Way Handshake Analysis

6. Inspect the `hostapd` foreground console debug log output on Access Point `<initials>-WAP0` captured during Host H1's reconnection:
   - Identify the log messages corresponding to the 4 steps of the **4-Way Handshake** (`sending 1/4 msg` , `2/4 Pairwise` , `sending 3/4 msg` , `4/4 Pairwise`). How many total EAPOL key messages are exchanged?
   - What type of keys (PTK, GTK, PMK) are derived and installed during this handshake sequence as shown in the log output?
   - Why is the 4-Way Handshake necessary (what does it protect against in a wireless environment)?

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8. Troubleshooting & Diagnostic Logging Commands

Real-Time Wireless Logging & Debugging Reference

1. Verbose Daemon Logging (`wpa_supplicant` & `hostapd`)

Run daemons in the foreground with extra debug flags (`-d` or `-dd`) to watch associations, probe requests, and 4-way handshakes live:

```
# On Client Nodes (H0 / H1) - Verbose Client Debugging:
pkill wpa_supplicant
wpa_supplicant -dd -i wlan0 -c /tmp/wpa_supplicant.conf

# Log client debug output to a file:
wpa_supplicant -dd -i wlan0 -c /tmp/wpa_supplicant.conf -f /tmp/wpa_debug.log &

# On Access Point (WAP0) - Verbose AP Debugging:
pkill hostapd
hostapd -dd /home/cs457/hostapd.conf
```

2. Real-Time Wireless Event Monitoring (`iw event`)

Monitor raw `nl80211` wireless events (auth, assoc, disassoc, connect, scan results) in real time:

```
# Monitor all kernel netlink wireless events live with timestamps:
iw event -t

# View associated station statistics (signal dBm, RX/TX bytes/packets):
iw dev wlan0 station dump
```

3. Real-Time Interactive CLI Debugging (`wpa_cli`)

Dynamically alter log levels or attach to the event log stream interactively:

```
# Increase log level dynamically without restarting daemon:
wpa_cli -i wlan0 log_level debug

# Open interactive wpa_cli monitoring shell:
wpa_cli -i wlan0
> level 0        # Shows all events (EAPOL, scan, state changes) live in terminal
> status         # Check current state detail
```

4. Live Console Packet Capture (`tcpdump`)

Inspect link-layer headers and EAPOL security frames directly on the Linux console:

```
# Capture and log 802.11/Ethernet link-layer headers live:
tcpdump -i wlan0 -e -n -v

# Filter specifically for EAPOL 4-Way Handshake frames (EtherType 0x888e):
tcpdump -i wlan0 -e -n -v 'ether proto 0x888e'

# Monitor ICMP ping traffic live:
tcpdump -i wlan0 -n icmp
```

5. Interface Errors & Kernel Logs (`dmesg` / `ip`)

```
# Check kernel wireless driver events:
dmesg | grep -E "wlan|mac80211|cfg80211|ath|iwl"

# Continuously follow kernel log output:
dmesg -w

# View detailed interface stats, dropped packets, and errors:
ip -s link show wlan0
```

Student Troubleshooting Notes

Issue Encountered: [Describe what went wrong, if anything]
Discovery Method: [How did you find the issue?]
Resolution: [How did you fix it?]

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9. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- The Wireshark packet capture file (`wlan_traffic.pcap` or `wlan_traffic.pcapng`) containing the captured 802.11 beacon frames and wireless traffic.
- A single PDF report containing:
  - Your written answers to the 6 Wireshark analysis questions in Section 7.
  - A screenshot of your active CML topology showing the running hosts (`<initials>-H0` , `<initials>-H1`) and AP (`<initials>-WAP0`).
  - A screenshot of Host H0's console showing the output of `iw dev wlan0 link` or `wpa_cli status` , confirming association.
  - A screenshot of the AP (`<initials>-WAP0`) console showing the output of `hostapd` running in foreground debug mode (`hostapd -dd /home/cs457/hostapd.conf`), confirming the full WPA2 4-Way Handshake sequence (`sending 1/4 msg` , `received EAPOL-Key frame (2/4 Pairwise)` , `sending 3/4 msg` , `received EAPOL-Key frame (4/4 Pairwise)` , and `EAPOL-4WAY-HS-COMPLETED`).